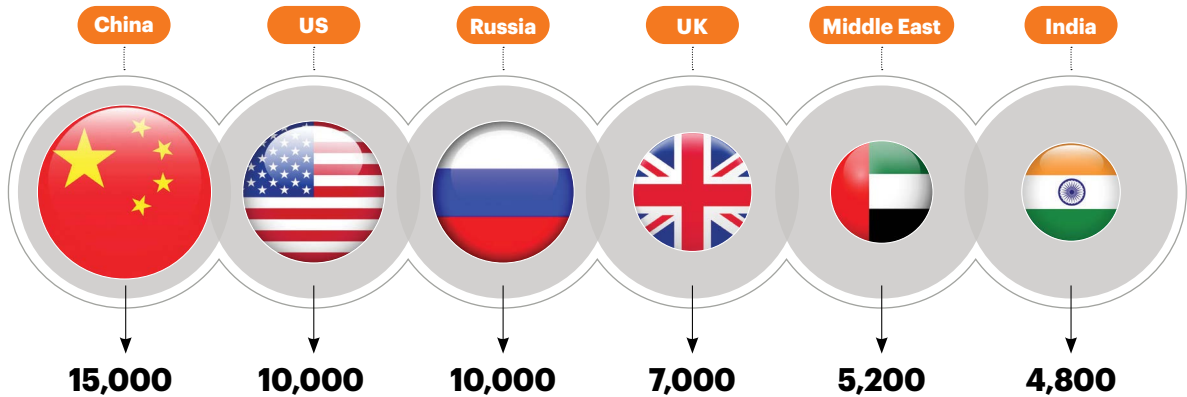


SMALL IS BIG

- By 2035, small modular reactors (SMRs) are poised to unlock a potential \$20 billion opportunity
- SMRs are expected to fill a critical gap in India's clean energy transition



NOTE CAPACITY POTENTIAL (MW) SOURCE ENINRAC RESEARCH, IAEA

Hence, operational expenditure of ₹47.47 lakh/MW (as per 2017-18 estimates) as given in the RFP is significantly on the higher side and makes the overall project unviable,” Adani Energy Solutions has said. In response, NPCIL has said that the operating and maintenance (O&M) expenditure indicated in the RFP has been taken from plants operating under similar conditions.

Naveen Ahlawat, Head of the Power to X vertical, Green Hydrogen, Green Steel and CCUS at Jindal Steel, says consultations are on with the government to make the plants viable. “The O&M cost proposed by NPCIL is very high. Our calculations show it should be 40-50% lower, ₹15-20 lakh/MW, as with the involvement of private players, it will be all about volumes,” he says.

Another concern is the proposed expertise fee of 60 paise per unit. The industry wants it reduced to 5-7 paise per unit. The companies say such a high cost will have a

negative impact on the company's commercial viability, make tariffs uncompetitive, and hinder development of the sector.

The annual energy produced from a twin BSR unit of 440 MW at 72.5% PLF (net of auxiliary consumption) would be around 2,500 MUs (mega units). The expertise charge of 60 paise/kWh on this will be around ₹150 crore annually. The present value of these yearly expertise charges, discounted over 40 years of operation at a 12% discount

rate, is approximately ₹1,250 crore. This cost is significant, especially when compared to the total project cost of ₹3,000-3,500 crore. “The expertise fee of 60 paise per unit of nuclear power charges by NPCIL is high and will cut down the returns from nuclear power. I think NPCIL is looking at it as a revenue stream,” says Ahlawat. NPCIL says the fee has been decided by the Centre.

LOW PLF

The industry has also flagged NPCIL's proposal to maintain the plant load factor (PLF), which shows the efficacy and profitability of a plant, for nuclear power at 65%, significantly lower than its own reactors, which typically reach 90-95%.

Reliance Industries has asked NPCIL to explain the methodology used to arrive at the PLF as the plant at Kaiga, using the same reactors, has been averaging above 90% every year over the last five years.

**100
GW**

**Nuclear power
generation
target by 2047**